SUMMARY · MK 2 FINANCIAL MANAGEMENT & STRATEGY · WEEK 3

Cash Flow & Financial Planning (Cash Budgets & Pro Forma)

After reading and analyzing the past, this week looks forward: measuring cash flow (depreciation, OCF, free cash flow), building a cash budget for short-term needs, and preparing pro forma statements. Based on Zutter & Smart ch. 4, applied in the Indofood 5-year forecast assignment.

Material

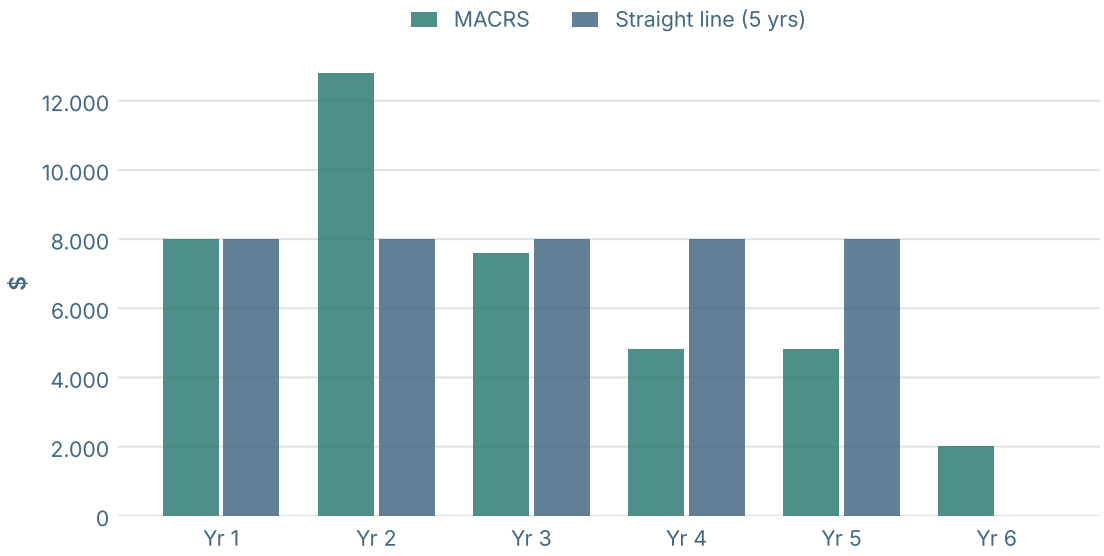
1 Depreciation: an expense that uses no cash

Depreciation spreads a fixed asset's cost over its life — a *noncash charge*. Depreciable value = cost + installation (Baker: $38k + $2k = $40k). US tax uses MACRS (5-year class: 20/32/19/12/12/5%). Indonesia's rules differ, but faster depreciation = lower early tax = more early cash.

2 Why faster depreciation means more cash early Deep dive

Depreciation creates a tax shield = depreciation × tax rate; MACRS front-loads it, which is worth more in present value.

Baker $40,000 machine: MACRS 5-year vs straight line




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Year 2: $12,800 × 21% = $2,688
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3 Measuring cash flow: sources vs uses, OCF, FCF

Sources	Uses
Asset decrease; liability increase; net profit; depreciation; stock sale	Asset increase; liability decrease; loss; dividends; buyback

Baker 2019: CFO 837, CFI −300, CFF +20 → cash +557. `OCF = EBIT(1−T) + Dep = 392.3` `FCF = OCF − NFAI − NCAI = 392.3 − 300 − 57 = 35.3` Positive but below $70 interest — not enough return for investors.

4 Cash budget: forecasting shortfalls and surpluses

A **cash budget** plans monthly receipts and disbursements, driven by the sales forecast (external or internal). Receipts − disbursements = net cash flow; + beginning cash = ending cash; vs minimum balance → required financing or excess cash.

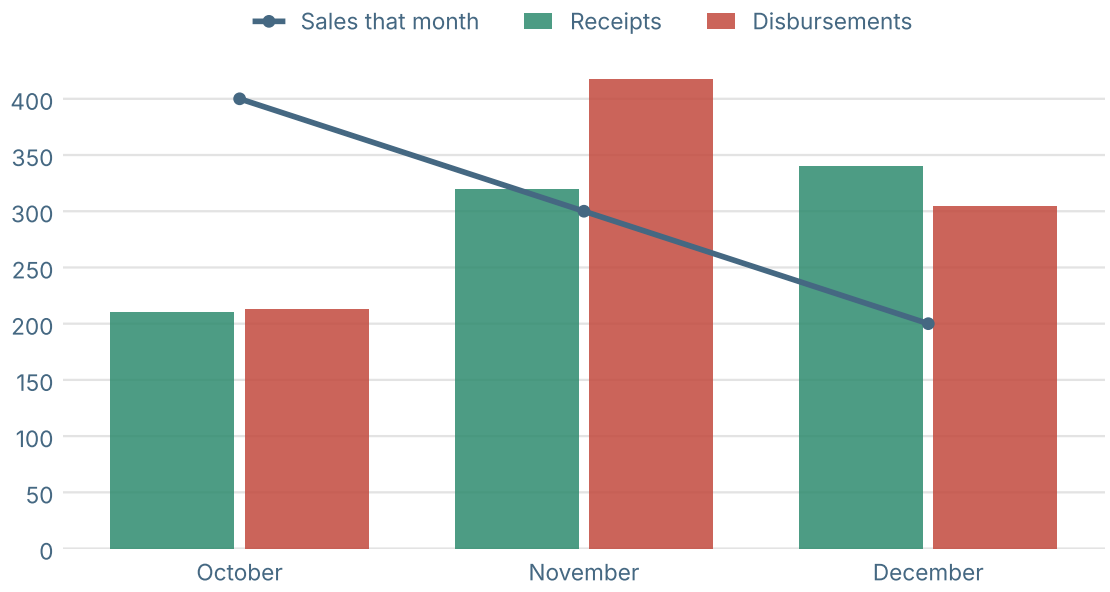
Coulson: Oct receipts 210/disbursements 213 → excess 22; Nov 320/418 → borrow 76; Dec 340/305 → repay to 41. Highest sales month (Oct) isn't the tightest cash month (Nov) — profit ≠ cash. Arrange credit lines in advance.

Scenario analysis: worst-case peak borrowing $190 vs most-likely $76. Monitor intra-month cash too. Same logic for personal budgets.

5 Coulson's cash budget charted: profit ≠ cash

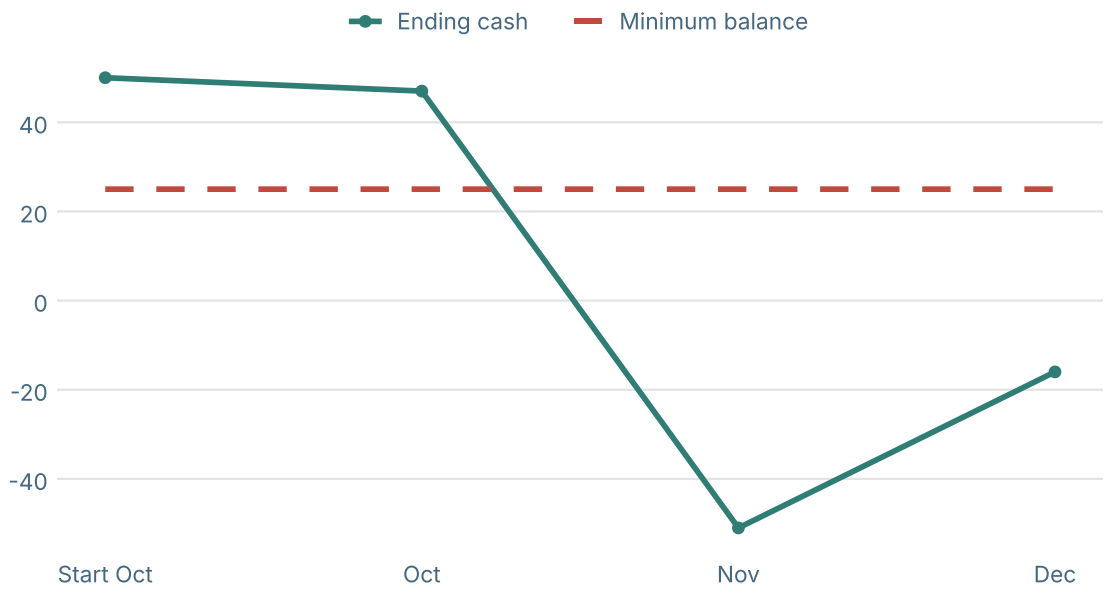
Deep dive

Coulson: cash receipts vs disbursements ($000)



Sales peak in October but cash is tightest in November (lagged collections, $130 machine).

Ending cash vs minimum balance



Plan for the worst case

Pessimistic peak borrowing: $190.

6 Pro forma statements: projected income and balance sheet

Fair Traders: sales $100k → $135k forecast.

Percent-of-sales income statement gives EBT $12,150 but assumes all costs variable. Splitting fixed/variable costs gives **$28,250** — operating leverage (see Week 5).

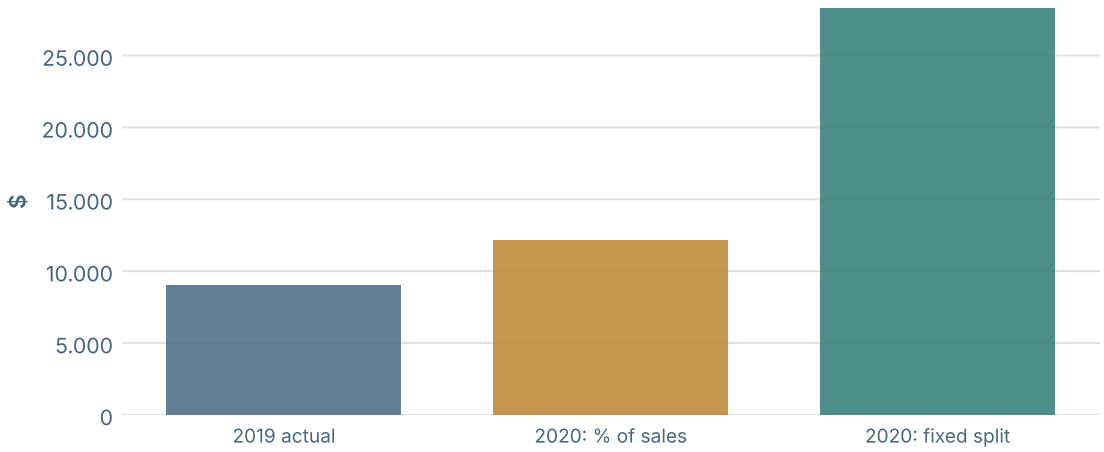
Judgmental balance sheet: estimate each line (cash $6k, receivables 45 days = $16,875, inventory $16k, net fixed assets $63k, payables 73 days = $8,100, taxes payable $455, retained earnings $29,327) → assets $105,875 vs funded $97,582 → **external financing required $8,293** (the plug).

Weaknesses: assumes the past repeats and forces some values; still valuable for projected ratios and adjusting plans.

7 Percent-of-sales vs fixed/variable split: a big difference

Deep dive

Fair Traders: projected 2020 EBT



Percent-of-sales gives $12,150 EBT; splitting fixed and variable costs gives $28,250 — operating leverage.

For the Indofood task

Separate truly variable costs (wheat, CPO, packaging) from fixed ones.

Key takeaways

1. Depreciation is a noncash charge; faster depreciation = lower early taxes.
2. Asset increase = cash out; liability increase = cash in.
3. OCF = EBIT(1−T) + Dep; FCF = OCF − NFAI − NCAI.
4. Cash budgets reveal borrowing needs in advance — arrange credit lines early.
5. Use scenarios — worst-case needs can far exceed the base case.
6. Percent-of-sales ignores fixed costs; split fixed vs variable.
7. External financing required is the balance-sheet plug.

Case studies

Case & worked answers

Group assignment: Indofood 5-year forecast (2026–2030)

Task: fill Indofood's assumptions (growth, operating cost ratios, non-operating items, tax & attribution, cash flow) and justify each versus the 2025 reference.

Links: sales growth is the key pro forma input; cost ratios = percent-of-sales (but not all costs are variable — wheat prices and the rupiah drive COGS; CPO affects agribusiness); working capital and capex = NCAI/NFAI logic; dividends must be cash-supported.

Pitfalls: optimistic growth with falling cost ratios unexplained; unsupported payouts; inconsistent capex and depreciation. Finally check margins, net income, cash and that the balance sheet balances.

Worked examples

Problem 1 Medium **OCF dan free cash flow**

EBIT Rp500 juta, pajak 25%, penyusutan Rp80 juta, investasi aset tetap bersih (NFAI) Rp150 juta, kenaikan modal kerja bersih (NCAI) Rp40 juta. Hitung OCF dan FCF.

1. NOPAT = EBIT × (1 − T) = 500 × 0,75 = 375.
2. `OCF = NOPAT + Penyusutan = 375 + 80 = 455`
3. `FCF = OCF − NFAI − NCAI = 455 − 150 − 40 = 265`

Answer: OCF Rp455 juta; FCF Rp265 juta (kas yang bebas dibagikan ke pemberi dana).

Problem 2 Medium **Garis lurus vs MACRS**

Mesin Rp100 juta, umur 5 tahun, tanpa nilai sisa. Tarif MACRS 5 tahun: 20%, 32%, 19,2%, 11,52%, 11,52%, 5,76%. Pajak 25%. Berapa penyusutan tahun 1 dan 2 dengan kedua metode, dan berapa pajak yang "tertunda" di tahun 2 bila memakai MACRS?

1. Garis lurus: 100 ÷ 5 = 20 per tahun → tahun 1 = 20, tahun 2 = 20.
2. MACRS: tahun 1 = 20% × 100 = 20; tahun 2 = 32% × 100 = 32.
3. Selisih penyusutan tahun 2 = 32 − 20 = 12.
4. Penghematan pajak = 12 × 25% = **Rp3 juta** di tahun 2. Total penyusutan tetap 100 — yang berubah hanya *waktunya*.

Answer: Tahun 2: MACRS 32 vs garis lurus 20 → pajak lebih rendah Rp3 juta (kas datang lebih awal).

Problem 3 Medium **Anggaran kas**

Penjualan Jan 100, Feb 120, Mar 150 (Rp juta). 30% tunai, 70% diterima bulan berikutnya. Pengeluaran Maret 140. Saldo kas awal Maret 20, saldo minimum 15. Perlu pinjam berapa?

1. Penerimaan Maret = 30% × penjualan Maret + 70% × penjualan Februari = 0,3 × 150 + 0,7 × 120 = 45 + 84 = 129.
2. Saldo akhir sebelum pinjaman = 20 + 129 − 140 = 9.
3. Kekurangan dari minimum = 15 − 9 = **6**.

Answer: Perlu pinjaman jangka pendek Rp6 juta di bulan Maret.

Practice questions

Try answering first. The answer key is below each question.

1. Depreciation is a noncash charge because...

A. It doesn't reduce profit
B. It reduces profit but no cash leaves that year
C. Paid in cash yearly
D. Balance sheet only

Answer: B. Cash left when the asset was bought.

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2. Coulson: November ending cash −51, minimum 25. Required financing...

A. 25
B. 51
C. 76
D. 98

Answer: C. From −51 to 25 = 76.

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3. OCF 392.3, NFAI 300, NCAI 57. FCF...

A. 35.3
B. 92.3
C. 449.3
D. −35.3

Answer: A. 35.3.

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4. Why can percent-of-sales understate profit when sales grow?

A. Ignores tax
B. Treats all costs as variable
C. Uses old prices
D. Profit can't be projected

Answer: B. Operating leverage.

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5. In a judgmental pro forma balance sheet, external financing required is...

A. Retained earnings
B. The balancing plug
C. Fixed assets
D. Revenue

Answer: B. Closes the gap.

Sources

- Materi Week 3 - MK02.pdf (Zutter & Smart, Principles of Managerial Finance 15e, Bab 4)
- Pre-Assignment Week 3 Instructions - Indofood Forecast
- Indofood_5Year_Forecast_Simple_Worksheet.xlsx